

Integration of Remote Floating Gate FET-based Biosensor with Millifluidic Systems For Point-Of-Care Diagnostics

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Methods of developing low-cost biosensors that are rapidly-producible and highly accurate are necessary for effective response to disease outbreaks. Early detection during disease outbreaks can reduce the long-term spread of disease, which reduces the loss of life, and prevents a shutdown of societal functions. Remote-floating gate field effect transistors (RFGFETs) can potentially improve population-based rapid detection capabilities during pandemics. RFGFETs possess many of the necessary qualities of an ideal sensor device to use in emergency scenarios, and have 3 key advantages over other methods of detection: (1) the limit of detection (LoD) is 11 magnitudes lower than the LoD of traditional at-home diagnostic devices, (2) RFGFETs provide results at the minute time-scale rather than hours or days, (3) RFGFETs are low-cost. However, the preparation method that precedes the usage of RFGFETs for pathogen detection is labor-intensive and must be completed in a laboratory setting. Here, we report a simplified procedure as an alternative to the traditional preparation method for RFGFET devices to widen the scope of its use beyond the laboratory setting. This will be accomplished by fabricating a millifluidic device to automate the preparation steps of the sensor platform. The overall preparation process was shortened from approximately eighteen hours to four hours, and concentrations of antigen in the ng/mL range were detected.

Millifluidics | Biosensors | FET sensors | RFGFET

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Introduction

Conventional detection methods for viral proteins such as the enzyme-linked immunosorbent assay (ELISA), and methods of detection for nucleic acids such as Polymerase Chain Reaction (PCR) can be time intensive and difficult to replicate outside of laboratory settings due to the need for laboratory equipment and specialized training (1–3). On the other hand, disease tracing programs implemented during the onset of a potential pandemic require rapid-detection and the ability to set up equipment in low-resource areas (4–9). The discrepancy between the time-intensive process required for accurate detection and the rapid response that is necessary during emergency situations causes spikes in the number of people that become infected during pandemics and epidemics (10). There exists a need for diagnostic testing that is both rapid and accurate.

A potential solution to the aforementioned challenges is a remote-floating gate field effect transistor (RFGFET) (11,

12). This novel device offers rapid, sensitive detection of viral proteins such as the SARS-CoV-2 spike protein, while having a limit of detection (LoD) that is 10,000-fold lower than typical ELISA protocols. However, one major challenge to increasing the use of RFGFETs is the time-intensive treatment process that is required to arrive at a fully-functional sensing platform. This treatment process requires several hours of incubation of antibody proteins, as well as an overnight incubation period. The final product also has a short shelf life, which further prevents RFGFETs from being used widely.

Fluidic devices can be incorporated into the production process of RFGFETs to simplify the production process. They can be used in low-resource settings with minimal equipment, and offer faster response times compared to the hours-long wait of a PCR or ELISA protocol (13–17). Furthermore, incorporating fluidic devices introduces convective flow into the preparation steps that involve incubation of a protein solution for several hours in static preparation schemes. Typically, incubation of a protein to immobilize it onto a surface relies only on passive diffusion, resulting in long incubation times. In a millifluidic device, reagents are introduced into the device using a syringe pump, and the flowing solutions can now immobilize onto the sensing surface as a result of convective flow, which results in shorter preparation times. Taken together, millifluidic devices can reduce the preparation time of RFGFETs so that their use in pathogen detection can be carried out on a population-wide basis during periods of emergency, such as during the onset of a pandemic.

Methods

A. Preparation of the Sensing Surface. In the traditional preparation method, the sensing surface is developed by treating a silica glass slide with 2% (3-Aminopropyl)triethoxysilane (APTES) in acetone for 30 seconds. (1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride) (EDC) and (N-hydroxysulfosuccinimide) (sulfo-NHS) are combined with the primary antibody (primary Abs), Goat Anti-mouse Immunoglobulins (IgGs) for 15 minutes, after which the pH of the solution is raised by treating the solution with sodium bicarbonate at a 1:4 dilution. This solution is applied to the silica glass slide for 1 hour. After washing with phosphate-buffered saline (PBS),

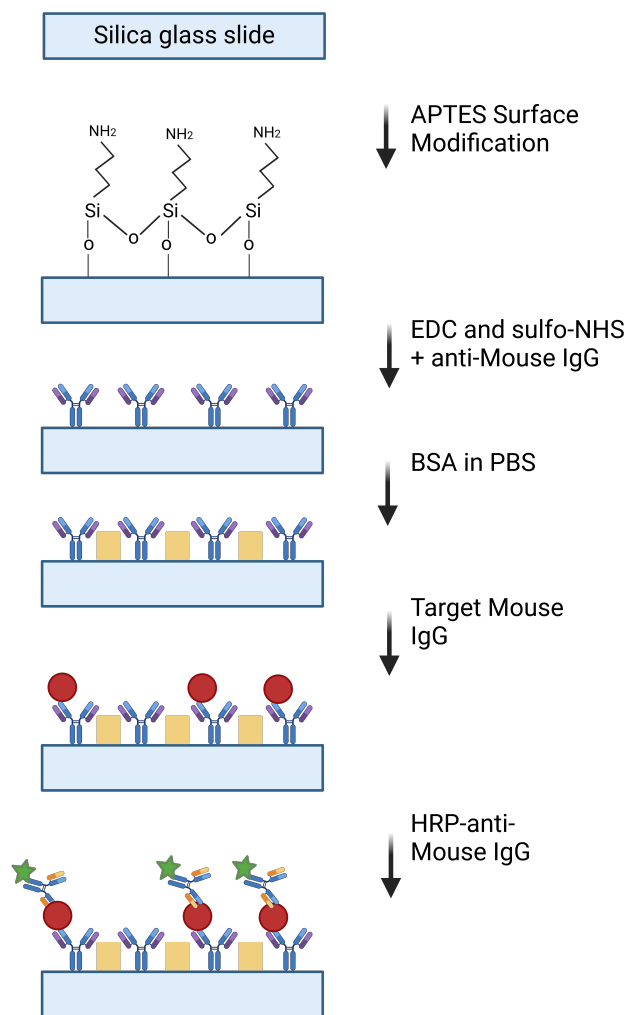


Fig. 1. Treatment of a silica glass slide as part of an APTES-based sandwich ELISA protocol. This treatment method is referred to as the stagnant, or static procedure.

a 5% w/v bovine serum albumin (BSA) in PBS solution is applied to the surface. In the traditional preparation method, the BSA in PBS solution is applied overnight, at 4°C. In the millifluidic device, a 5% w/v solution of BSA is prepared in PBS containing 0.05% v/v Polysorbate 20 (PBS-T). The blocking layer is rinsed with PBS-T. A solution of Mouse IgGs at the desired concentration is added to the surface and incubated for one hour. The antigen solution is rinsed with PBS-T, and a solution of 2 µg/mL of the secondary Goat anti-Mouse IgG (sAbs) is applied to the surface and incubated at room temperature for one hour. The sAb is conjugated with horseradish peroxidase (HRP).

B. Electrical Measurement Setup. Copper tape and indium-tin oxide (ITO) remote floating gate field effect transistors (RFGFETs) were clamped to the surface of the glass slides prior to electrical measurement. A solution of fluorophenol and 5.55 mM hydrogen peroxide in a ratio of 1:9 by volume is drop-casted onto the surface, and the transfer curves were measured for 100 cycles using a Kiethley 4200A semiconductor analyzer, as shown in Figure 2. All transfer curves were set to a 50 mV drain voltage, and the gate volt-

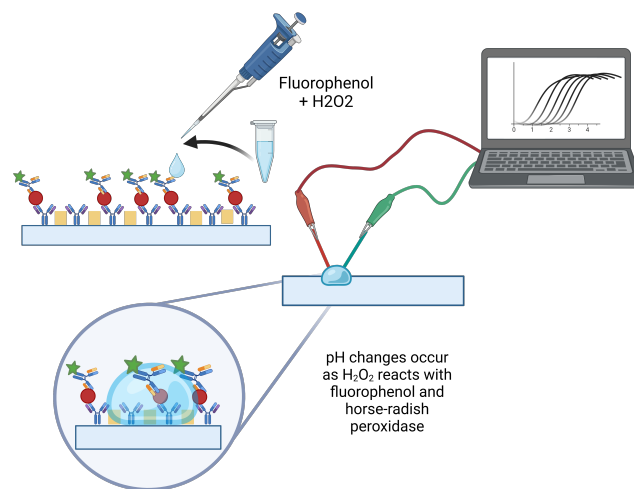


Fig. 2. Preparation of treated glass slide for testing

age remained in the double-sweep mode.

C. Preparation of the Millifluidic Device. The millifluidic device features 3 straight channels, with a length of 20.3 mm, and a width and depth of 1 mm. The device was 3D-printed on a FormLabs Form 2 Stereo-lithography (SLA) printer, using the FormLabs Clear Resin V4, with a layer depth of 0.05 mm. Inlet and outlet ports are incorporated onto the top surface of the device. The bottom surface of the device features the 3 fluidic channels. The assembly is layered in order, with the acrylic adhesive sandwiched between the fluidic piece and the glass substrate, and then clamped for 15 minutes to allow for sufficient adhesion prior to usage. The 3D-printed piece is attached to the APTES-treated glass substrate using a high-strength acrylic adhesive (Figure 4). The adhesive has rectangular channels cut out using a cutter plotter to allow fluids passing through the 3D-printed piece to contact the glass substrate. Silicone tubing with an inner diameter of 1/16" connects the millifluidic device to the syringe and syringe pumps.

D. Millifluidic Device Usage Procedure. The first reagent that is passed through the millifluidic device is the EDC, sulfo-NHS, and primary antibody solution, which has been diluted with sodium bicarbonate. This solution is initially flowed through at 20 µL/min using the New Era Pump Systems NE-300 Syringe Pump, until the entire channel is saturated with the solution. The solution is allowed to remain flowing over the glass surface for 20 minutes, after which it remains saturated on the surface for 40 minutes. A syringe is then used to extract the solution and the channels are rinsed twice with PBS-T.

The rinsing procedure involves PBS-T being passed through the device at 55 µL/min until the solution saturates the entire channel. The PBS-T is then removed from the device using a syringe connected to the outlet of each channel, and more PBS-T is passed through the channel at the same flow rate, and again removed via syringe. This rinsing process is repeated in between each treatment step.

The blocking solution, a 5% BSA in PBS-T solution, is

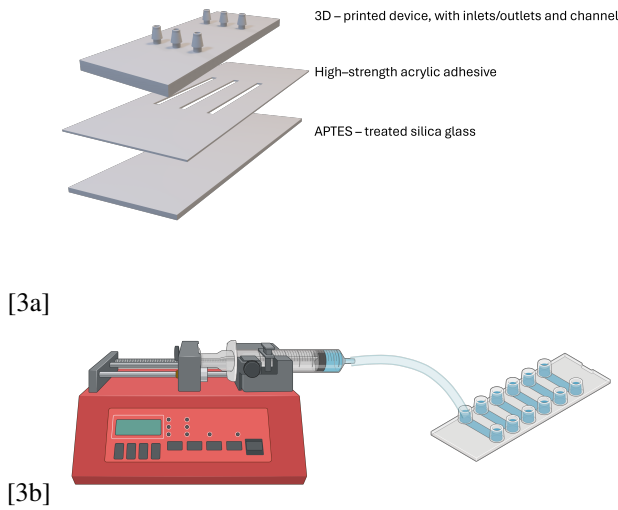


Fig. 3. (a) The three layers of the millifluidic device include the 3D-printed piece, which is attached to the glass substrate using an acrylic adhesive. (b) Silicone tubes are used to connect syringes to the inlets of the device, with the syringes positioned inside of a syringe pump.

passed through the device at a flow rate of 20 $\mu\text{L}/\text{min}$, and allowed to flow over the surface for 20 minutes. The formation of the blocking layer is typically the longest step in the static preparation method, where it is usually allowed to incubate between four to twenty-four hours. After removing the remaining blocking solution that was incubating inside the millifluidic channels, the channels are rinsed twice with PBS-T.

An antigen solution of Anti-Mouse IgG is prepared at the desired concentration and flowed through the millifluidic device at a flow rate of 20 $\mu\text{L}/\text{min}$. It is then rinsed with PBS-T twice and then the sAb, the HRP-labelled mouse IgG, is passed through the device at a flow rate of 20 $\mu\text{L}/\text{min}$ and allowed to pass over the glass surface for 20 minutes. A final rinsing step is completed, after which the millifluidic top piece is removed from the glass substrate. The surface is now ready for testing.

Flow Simulation

In order to predict the velocity and pressure gradient of the fluidic channels, Ansys Fluent was used to simulate the flow of PBS through the device at a flow rate of 80 $\mu\text{L}/\text{min}$. 80 $\mu\text{L}/\text{min}$ was chosen as the flow rate for the simulations because it greatly exceeds the flow rates that are used for preparing the sensing surface. This ensures that the pressure throughout the device would not exceed the burst pressure of the millifluidic channels. The approximate burst pressure was calculated using the Lamé equation for thick walled pipes:

$$\text{Burst Pressure} = \frac{TS(OD^2 - ID^2)}{OD^2 + ID^2} \quad (1)$$

TS is the tensile strength of the millifluidic device material, which is 65 MPa for the FormLabs Clear Resin V4, OD is

taken as the wall thickness, and ID is taken to be the width of the rectangular channels. A burst pressure of 56.36 MPa was calculated for the system. The Ansys Fluent simulations show that the static pressure does not exceed 32 Pa, shown in Figure 4. In order to detect flow anomalies within the mil-

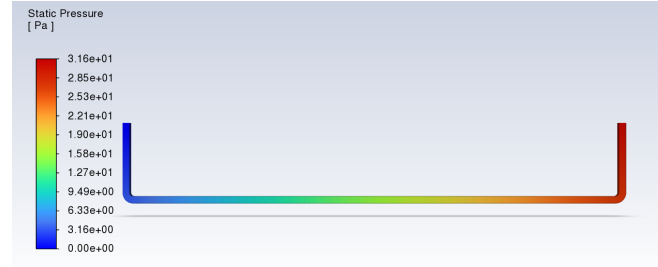


Fig. 4. Computational fluid dynamics software was used to produce the expected pressure gradient across the fluidic channel.

lfluidic channels, a velocity simulation was produced using Ansys Fluent. Monoclonal antibodies, such as the first solution that is passed through the millifluidic device, are susceptible to damage due to shear stress (18). The velocity simulation was used along with calculations of the shear rate to determine whether there was a high enough shear stress to cause protein degradation or aggregation in the silicone tube that was used to connect the syringe to the millifluidic device inlets, or within the channels. The average shear rate was calculated using Equation 2:

$$\bar{\gamma} = \frac{8Q}{3\pi r^3} \quad (2)$$

where Q is the volumetric flow rate and r is the inner radius. The shear rates were calculated for the solution passing through the silicone tubes and through the channels at 55 $\mu\text{L}/\text{min}$ and 20 $\mu\text{L}/\text{min}$. For shear rate calculations of fluid passing through the millifluidic channel, r in Equation 2 is taken as the channel width.

	$Q = 20 \mu\text{L}/\text{min}$	$Q = 55 \mu\text{L}/\text{min}$
Silicone Tube	258 s^{-1}	710 s^{-1}
Channel	1018 s^{-1}	2801 s^{-1}

Table 1. The shear rates for the two geometries and the two volumetric flow rates were calculated. The silicone tube has an inner diameter of 1.58 mm and the channel has a width of 1 mm.

It was found that the shear rates produced by the flow in the silicone tube and the channels did not exceed the shear rate of 10^4 s^{-1} , which has been investigated previously as a maximal value below which protein damage or unfolding does not occur (19).

Results

Prior to experimentation of the millifluidic device on the APTES-treated silica glass, control experiments were conducted to determine the viability of silica glass as the sensing surface substrate. These tests confirmed the presence of bound antigens on the silica surface in a static treatment method (Figure 5), indicating that it should be possible to immobilize primary Abs, form a blocking layer, and bind

Antigen Concentration (ng/mL)	ΔV	Standard Error
0	0.03	0.05
15	0.12	0.003
150	0.25	0.005
1150	0.28	0.05
5000	0.37	—

Table 2. The shear rates for the two geometries and the two volumetric flow rates were calculated. The silicone tube has an inner diameter of 1.58 mm and the channel has a width of 1 mm.

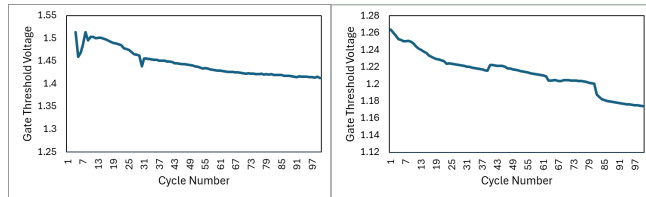


Fig. 5. Control experiments were conducted without flowing solutions over the glass substrates. The drop in the gate threshold voltage indicates the presence of antigens bound to the glass surface.

antigens and sAbs in a flow regime using a millifluidic device. In order to gain an idea of the relationship between the antigen concentration and the resulting signal that would be produced during the electrical measurement, a performance metric must be tracked across the different concentrations. The preparation process using the millifluidic device shows consistent changes in voltage drops for each concentration, making ΔV an appropriate metric to measure. The voltage drop for each concentration is related to the change in pH that occurs on the sensing surface while the 100 transfer curves are being measured using the semiconductor analyzer. When the fluorophenol and hydrogen peroxide solution comes into contact with the HRP tag on the sAbs, an decrease in pH is observed, due to the increased formation of H^+ ions. Therefore, a higher concentration of bound HRP-labelled sAbs will result in higher concentrations of H^+ , leading to a larger voltage drop across the transfer curves. The average voltage drop between V_{th} for the initial cycles and the final cycles is shown in table 2, along with the 95% confidence intervals. As expected, increasing the antigen concentration results in a greater ΔV .

A preliminary fitting of the data to a equation that relates ΔV to the antigen concentration was completed (Figure 6).

An in-depth characterization of the millifluidic device, which includes testing a wider range of antigen concentration, would be necessary to develop a comprehensive model fit. The data collected shows a linear relationship between ΔV and antigen concentration at low concentrations, which flattens out to a near-constant value at higher concentrations. This is typical of a Langmuir-isotherm model, which was used to develop an equation for ΔV and the antigen concentration:

$$\Delta V = \Delta V_{max} \frac{K_a c}{1 + K_a c}$$

Where c is the antigen concentration, K_a is the binding constant of antigens onto the antibodies, and ΔV_{max} represents the maximum change in the surface potential that can be mea-

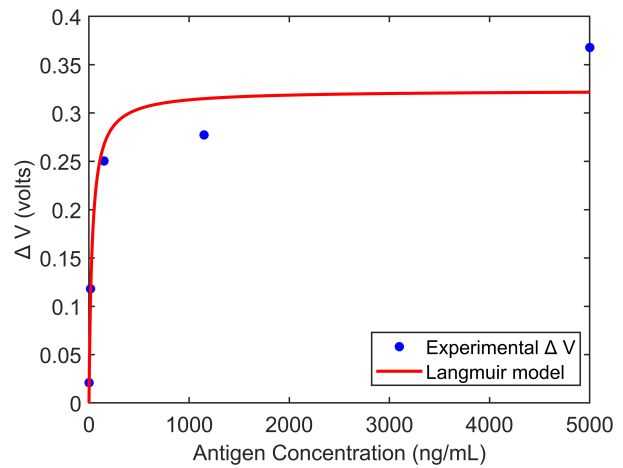


Fig. 6. The experimental data can be fitted to a Langmuir isotherm model to relate the change in voltage and the antigen concentration, with an R^2 value of 0.9405

sured by the sensor device. Matlab 2024a Curve Fitting Toolbox was used to fit the data to a Langmuir-isotherm model to find values for ΔV_{max} and K_a that resulted in the highest value of R^2 . ΔV_{max} was 0.32 V, the value of K_a was found to be $5.3 \times 10^{-9} M^{-1}$. K_d , the dissociation constant, is therefore $1.9 \times 10^{-10} M$, which is within the typical range for the binding of Goat anti-Mouse IgGs to Mouse IgGs.

Conclusions

In conclusion, this work shows that it is possible for millifluidic devices to be utilized in the preparation of pathogen-sensing surfaces using RFGFETs. The millifluidic device decreases the length of time necessary for the preparation of the sensing surface, and further work should be completed to characterize the performance of the millifluidic device, to investigate whether detection of lower concentrations of antigen can occur. The data has shown ΔV for a specific concentration can be reproduced, a preliminary indication that the sensor device is reliable. Control experiments were conducted prior to the assembly of the millifluidic device to confirm that the traditional preparation process could be replicated on APTES-treated silica glass. Flow experiments were conducted to find flow rates and values for the duration of fluid flow that would promote adsorption and binding of proteins. Computational fluid dynamics was used to investigate the stability of the materials used in the millifluidic device and to avoid protein damage during operation of the device. The advantages to using the device design described in this paper are the ease of construction, as well as the time-efficiency of the preparation method. Additionally, the device can be disassembled by hand after all solutions are flowed through the device so that it can be used with the electrical measurement system. Unlike polydimethylsiloxane (PDMS) fluidic devices, which often necessitate the use of a clean room, these devices can be made easily using a conventional SLA printer, allowing for rapid-prototyping. It is also possible to manufacture the 3D-printed layer in large batches, and the 3-layer device, consisting of the 3D-printed piece,

the acrylic adhesive, and the glass substrate, requires minimal processing time.

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Bibliography

1. N. Younes, D. W. Al-Sadeq, H. Al-Jighefee, S. Younes, O. Al-Jamal, H. I. Daas, H. M. Yassine, and G. K. Nasrallah. Challenges in laboratory diagnosis of the novel coronavirus sars-cov-2. *Viruses*, 12(6):582, 2020. doi: 10.3390/v12060582.
2. Jameel M Abduljalil. Laboratory diagnosis of sars-cov-2: available approaches and limitations. *New microbes and new infections*, 36:100713, 2020.
3. A Cassidy, A Parle-McDermott, and RJFIMB O'Kennedy. Virus detection: a review of the current and emerging molecular and immunological methods. *Frontiers in Molecular Biosciences*, 8:637559, 2021.
4. Avinash Premraj, Abi George Aleyas, Binitha Nautiyal, and Thaha J Rasool. Nucleic acid and immunological diagnostics for sars-cov-2: Processes, platforms and pitfalls. *Diagnostics*, 10(11):866, 2020.
5. Solomon Abebe Yimer, Birgitte Boonstra Booi, Gwen Tobert, Andrew Hebbeler, Paul Oloo, Polina Brangel, Maïna L'Azou Jackson, Richard Jarman, Danielle Craig, Michael Selorm Avumegah, et al. Rapid diagnostic test: a critical need for outbreak preparedness and response for high priority pathogens. *BMJ Global Health*, 9(4):e014386, 2024.
6. Søren Hansen and Ahmed Abd El Wahed. Point-of-care or point-of-need diagnostic tests: time to change outbreak investigation and pathogen detection. *Tropical Medicine and Infectious Disease*, 5(4):151, 2020.
7. Quinn Matthews, Severino Jefferson Ribeiro da Silva, Masoud Norouzi, Lindomar José Pena, and Keith Pardee. Adaptive, diverse and de-centralized diagnostics are key to the future of outbreak response. *BMC biology*, 18:1–5, 2020.
8. Aliu Olalekan Olatunji, Janet Aderonke Olaboye, Chukwudi Cosmos Maha, Tolulope Olagoke Kolawole, and Samira Abdul. Revolutionizing infectious disease management in low-resource settings: The impact of rapid diagnostic technologies and portable devices. *International Journal of Applied Research in Social Sciences*, 6(7):1417–1432, 2024.
9. Gerald J Kost, William Ferguson, Anh-Thu Truong, Jackie Hoe, Daisy Prom, Arirat Banpavichit, and Surin Kongpila. Molecular detection and point-of-care testing in ebola virus disease and other threats: a new global public health framework to stop outbreaks. *Expert review of molecular diagnostics*, 15(10):1245–1259, 2015.
10. Pierre Nouvellet, Tini Garske, Harriet L Mills, Gemma Nedjati-Gilani, Wes Hinsley, Isobel M Blake, Maria D Van Kerkhove, Anne Cori, Ilaria Dorigatti, Thibaut Jombart, et al. The role of rapid diagnostics in managing ebola epidemics. *Nature*, 528(7580):S109–S116, 2015.
11. Hyun-June Jang, Xiaoyu Sui, Wen Zhuang, Xiaodan Huang, Min Chen, Xiaolei Cai, Yale Wang, Byunghoon Ryu, Haihui Pu, Nicholas Ankenbruck, et al. Remote floating-gate field-effect transistor with 2-dimensional reduced graphene oxide sensing layer for reliable detection of sars-cov-2 spike proteins. *ACS applied materials & interfaces*, 14(21):24187–24196, 2022.
12. Hyun-June Jang, Wen Zhuang, Xiaoyu Sui, Byunghoon Ryu, Xiaodan Huang, Min Chen, Xiaolei Cai, Haihui Pu, Kathleen Beavis, Jun Huang, et al. Rapid, sensitive, label-free electrical detection of sars-cov-2 in nasal swab samples. *ACS applied materials & interfaces*, 15(12):15195–15202, 2023.
13. Hanliang Zhu, Zdenka Fohlerová, Jan Pekárek, Evgenia Basova, and Pavel Neužil. Recent advances in lab-on-a-chip technologies for viral diagnosis. *Biosensors and Bioelectronics*, 153:112041, 2020.
14. P Maffert, S Reverchon, W Nasser, C Rozand, and H Abaibou. New nucleic acid testing devices to diagnose infectious diseases in resource-limited settings. *European Journal of Clinical Microbiology & Infectious Diseases*, 36:1717–1731, 2017.
15. Shikha Sharma, Julia Zapatero-Rodríguez, Pedro Estrela, and Richard O'Kennedy. Point-of-care diagnostics in low resource settings: present status and future role of microfluidics. *Biosensors*, 5(3):577–601, 2015.
16. Jorge Ricardo Mejia-Salazar, Kamilla Rodrigues Cruz, Elsa Maria Materon Vasques, and Osvaldo Novais de Oliveira Jr. Microfluidic point-of-care devices: New trends and future prospects for ehealth diagnostics. *Sensors*, 20(7):1951, 2020.
17. Shih-Mo Yang, Shuangsong Lv, Wenjun Zhang, and Yubao Cui. Microfluidic point-of-care (poc) devices in early diagnosis: A review of opportunities and challenges. *Sensors*, 22(4):1620, 2022.
18. Camilla Moino, Fiora Artusio, and Roberto Pisano. Shear stress as a driver of degradation for protein-based therapeutics: More accomplice than culprit. *International Journal of Pharmaceutics*, page 123679, 2023.
19. Lukas Brückl, Thomas Schröder, Stefan Scheler, Rainer Hahn, and Corinna Sonderegger. The effect of shear on the structural conformation of rhg and igg1 in free solution. *Journal of Pharmaceutical Sciences*, 105(6):1810–1818, 2016.